STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 1

INTRODUCTION TO STATISTICS

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INTRODUCTION TO STATISTICS

STATISTICS *Statistics is a Science used to collect, organize, present, analyze and interpret data to make decisions.*

TYPES OF STATISTICS Basically there are two types of statistics.

1. Descriptive Statistics 2. Inferential Statistics

DESCRIPTIVE STATISTICS Descriptive statistics consists of methods used to organize, preset and summarize data by using tables, graphs and summary sheets.

INFERENTIAL STATISTICS Inferential statistics consists of methods used to find out something about a population based on a sample.

IMPORTANCE OF STATISTICS

1. Statistics is used in different fields of Sciences like Social Sciences, Plant Sciences, Physical Sciences and Medical sciences.
2. It plays an important role in planning and helps the government in formulating the policies.
3. It plays an important role in Business, industry and Banking sector.
4. statistics present the data in comparable form so it helps us comparing past results with present.
5. Forecasting is the main objective of statistics. Forecasting means estimating future values with the help of past data.

BASIC TERMS IN STATISTICS

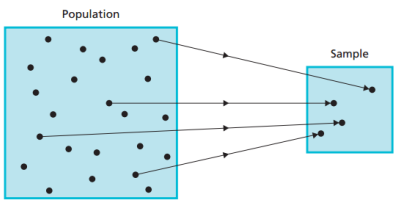
POPULATION A collection of all possible individuals, objects or measurements of interest is called a population. For example set of students in a college.

SAMPLE A portion or part of a population of interest is called a sample. For example set of students of Ist year class from a University.

PARAMETER Any numerical; value computed from the population is called a parameter. Parameters are fixed quantities.

STATISTIC Any numerical value computed from the sample is called a statistic. Statistics are variable quantities because they vary from sample to sample.

FIGURE 1.1 Relationship between population and sample



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DATA AND TYPES OF DATA

DATA Numerical facts or observations are collectively called data. A single numerical fact is called datum. There are different types of data.

UNIVARIATE DATA Data that represents observations on one variable that are measured in the same units is called Univariate data.

BIVARIATE DATA Data that represents observations on two variables that are measured in the different units is called Bivariate data.

MUTIVARIATE DATA Data that represents observations on more than two variables that are measured in the different units is called Multivariate data.

PRIMARY DATA Data that have been originally collected and have not undergone any sort of statistical treatment is called a primary data. For example marks of all the students in a class.

SECONDARY DATA Data that have undergone any sort of statistical treatment is called secondary data. For example percentage marks of each student in the class.

SOURCES OF PRIMARY DATA There are different sources of Primary Data.

- Through investigation
- Through Questionnaire
- Through Local Sources
- Through Telephone
- Through Internet

SOURCES OF SECONDARY DATA There are different sources of Secondary Data.

- Government Organizations
- Semi-government Organizations
- Research Organizations
- Newspapers
- Internet

CROSS-SECTION VERSUS TIME-SERIES DATA Based on the time over which they are collected, data can be classified as either cross-section or time-series data.

CROSS-SECTION DATA Data collected on different elements at the same point in time or for the same period of time are called cross-section data.

TIME-SERIES DATA Data collected on the same element for the same variable at different points in time or for different periods of time are called time-series data.

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VARIABLE AND TYPES OF VARIABLES

VARIABLE A variable is a characteristic under study that assumes different values for different elements. In contrast to a variable, constant assumes only one value ie its value is fixed.

Types of Variables Basically there are two types of variables.

1. Qualitative Variables 2. Quantitative Variables

QUALITATIVE VARIABLE A variable that can not be measured numerically but can be classified into two or more nonnumeric categories is called a qualitative variable or categorical variable or an attribute. The data collected on qualitative variable is called qualitative data. For example gender, religious affiliation and eye color etc.

QUANTITATIVE VARIABLE A variable that can be measured numerically is called a quantitative data. The data collected on a quantitative variable is called quantitative data. For example Incomes, heights, gross sales, prices of homes, number of accidents etc.

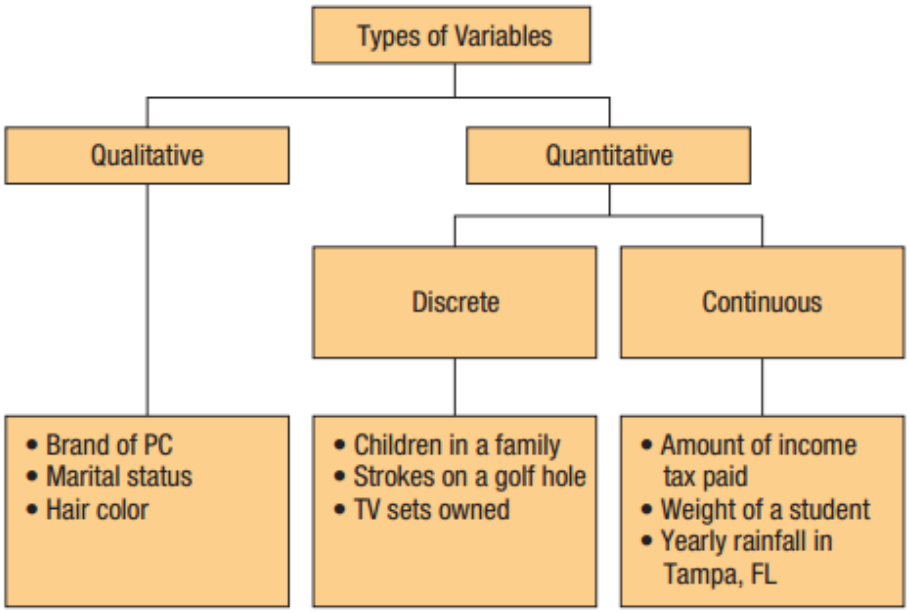
There are two types of quantitative variables

1. Discrete Variable 2. Continuous Variable

DISCRETE VARIABLE A variable is said to be discrete if it can assume only certain values in a given interval. These values are countable.
For example Number of bedrooms in a house, Number of doctors in a hospital etc

CONTINUOUS VARIABLE A variable is said to be continuous if it can assume any value in a given interval. These values are not countable.
For example temperature at a place, height of a person, speedometer etc

FIGURE 1.2 Summary of the types of variables



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MEASUREMENT SCALES OF VARIABLES

Measurement means assigning numbers to observations or objects and scaling is a process of measuring. Basically there are four scales of measurement w.r.t any data set

1. Ratio Scale
2. Interval scale
3. Ordinal Scale
4. Nominal Scale

RATIO SCALE Suppose that X is a variable taking two values $x_1 \; and \; x_2$. If it satisfy the following properties it can be referred as ratio scale.

$P-1$: Ratio is meaningful ie x_2/x_1 is meaningful.
$P-2$: Distance is meaningful ie $x_2 - x_1$ is meaningful.
$P-3$: There exists natural ordering. ie $x_2 \geq x_1$

EXAMPLES of ratio scale are Age, Weight, Height, Time, salary, Distance.

The Ratio scale data has the following properties.

1. Data categories are mutually exclusive and exhaustive.
2. Data categories are scaled according to the amount of the characteristic they possess.
3. Equal difference in the characteristic are represented by equal difference ij the numbers assigned to the categories.
4. The point 0 reflects the absence of the characteristic.
5. Ratio scale of measurement has a true zero point as its origin.

INTERVAL SCALE Suppose that X is a variable taking two values $x_1 \; and \; x_2$. If it satisfy the 2nd and 3rd properties of ratio scale than it can be referred as Interval scale.

$P-1$: Distance is meaningful ie $x_2 - x_1$ is meaningful.
$P-2$: There exists natural ordering. ie $x_2 \geq x_1$

EXAMPLES of interval scale are Temperature, IQ Score

A measurement scale possessing a constant interval size but not a true zero point, is called an interval scale.

The Interval scale data has the following properties.

1. Data categories are mutually exclusive and exhaustive.
2. Data categories are scaled according to the amount of the characteristic they possess.
3. Equal difference in the characteristic are represented by equal difference in the numbers assigned to the categories.
4. Interval scale of measurement does not have a true zero point.

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ORDINAL SCALE Suppose that X is a variable taking two values x_1 *and* x_2. If it satisfy only 3rd property of ratio scale than it can be referred as Ordinal scale.
$P-1$: There exists natural ordering. ie $x_2 \geq x_1$

EXAMPLES of ordinal scale are
(1) Grades of students *(A,B,C,D,E,F)*
(2) Ranking of Universities *(1st, 2nd, 3rd)*
(3) Socio Economic Status (*Poor, Middle class, Rich*)

The ordinal scale data has the following properties.

1. Data categories are mutually exclusive and exhaustive.
2. Data categories have a logical order.

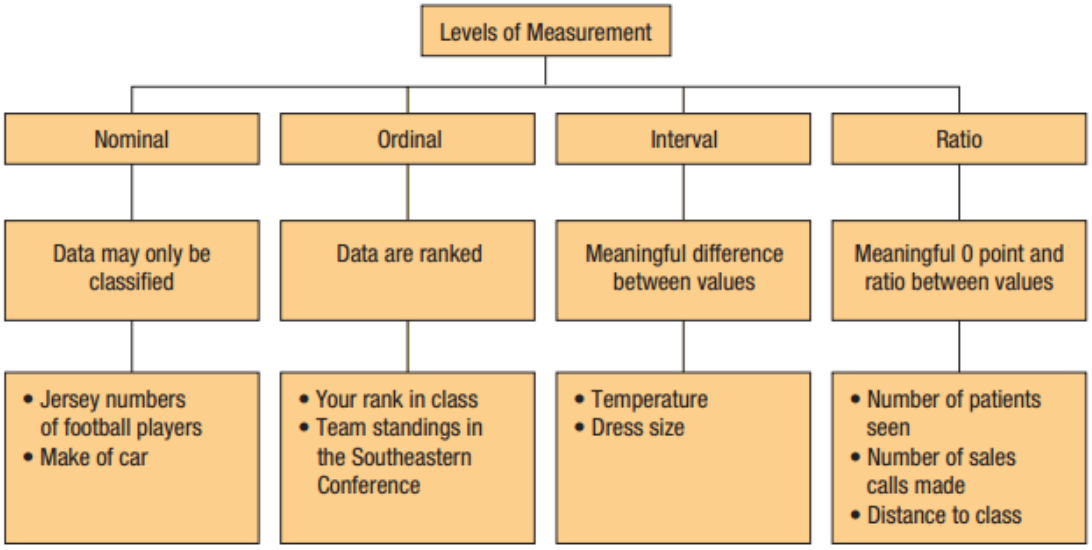
NOMINAL SCALE Nominal scale does not exhibit any property of ratio scale. It is a categorical variable,

EXAMPLES of Nominal scale are
(1) Gender
(2) Religion
(3) Nationality
(4) Eye color
(5) Marital Status

The nominal scale data has the following properties.

1. Data categories are mutually exclusive.
2. Data categories have no logical order.

FIGURE 1.3 Summary and Examples of the Characteristics for Levels of Measurement



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IMPORTANT NOTATIONS

SUM $\left(\sum\right)$ It is a Greek letter (read as sigma) and is used as a short-hand notation for sum.

In general $$y_1 + y_2 + ... + y_n = \sum_{i=1}^{i=n} y_i$$

For example if $y_1 = 2$, $y_2 = 3$ and $y_3 = 5$

then $\sum_{i=1}^{i=3} y_i = y_1 + y_2 + y_3 = 2 + 3 + 5 = 10$

PRODUCT $\left(\prod\right)$ It is a Greek letter (read as pi) and is used as a short hand notation for product.

In general $$y_1 \times y_2 \times ... \times y_n = \prod_{i=1}^{i=n} y_i$$

For example if $y_1 = 2$, $y_2 = 3$ and $y_3 = 5$

then $\prod_{i=1}^{i=n} y_i = y_1 \times y_2 \times y_3 = 2 \times 3 \times 5 = 30$

SIGNIFICANT FIGURES Those figures in a number which give its information other than its magnitude are called significant figures.

EXAMPLES

1. 26.2 contains 3 significant figures.
2. 1225 contains 4 significant figures.
3. 0.95 contains 2 significant figures.
4. 0.0037 contains 2 significant figures.
5. 0.001592 contains 4 significant figures.

ROUNDING OF FIGURES In order to express figures to smaller number of significant figures we usually round off the numbers. In rounding off the figures we examine the last significant figure. So

1. If it is less than 5 last significant figure will remain the same.
2. If it is greater than 5 than last significant figure is increased by 1.
3. If it is exactly 5 then
   (a) Increase the last significant figure by one if it is an odd number.
   (b) The last significant figure remains same if it is an even number.

EXAMPLES

1. 523 becomes 520
2. 526 becomes 530
3. 1975 becomes 1980
4. 1962 becomes 1962
5. 4.345 becomes 4.32

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SAMPLING TECHNIQUES

If the information you need is not already available from a previous study, you might conduct a **census**- i.e. by obtaining information for the entire population of interest. However, conducting a census may be time consuming, costly, impractical, or even impossible. Two methods other than a census for obtaining information are **sampling** and **experimentation**. We shall introduce the basics of both concepts of both methods.

SAMPLING TECHNIQUES

There are four major types of sampling Techniques

PROBABILITY SAMPLING

Sampling technique in which the elements to be selected are drawn on the basis of probability. Each element in a population has a certain probability of being selected as part of the sample. Probability samples allow you to make inferences about the population of interest.

There are four major types of probability sampling. These types are as follows

- Simple Random Sampling,
- Stratified Random Sampling
- Systematic Sampling
- Cluster Sampling

These sampling methods vary in their cost, accuracy, and complexity.

SIMPLE RANDOM SAMPLING *A sample selected in such a way that every element in the population or sampling frame has an equal probability of being chosen. Equivalently, all samples of size n have an equal chance of being selected.*

To select a simple random sample, first assign an identifying number to each element in the sampling frame. This is usually done sequentially using the same number of digits for each element. Then using random numbers with the same number of digits, select as many numbers as are needed for the sample size desired. Each numbered element in the sampling frame that corresponds to a selected random number is chosen for the sample. A sample selected using simple random sampling is usually denoted by **SRS**.

STRATIFIED RANDOM SAMPLING *In stratified random sampling, the population is divided into subgroups called* ***strata****. Then an SRS is selected from each stratum, and the SRSs in the strata are selected independently.*

The strata are often subgroups of interest to the investigator—for example, the strata might be different regions of the country in a survey of people. Elements in the same stratum often tend to be more similar than randomly selected elements from the whole population, so stratification often increases precision.

SYSTEMATIC SAMPLING *A sample in which every kth item of the sampling frame is selected, starting from a first element, which is randomly selected from the first k elements.*

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In systematic sampling, a starting point is chosen from a list of population members using a random number. That unit, and every kth unit thereafter, is chosen to be in the sample. A systematic sample thus consists of units that are equally spaced in the list.

CLUSTER SAMPLING *A sample obtained by stratifying the population, or sampling frame, and then selecting some or all of the items from some, but not all, of the strata.*

The cluster sample is a multistage design. It uses either random or systematic methods to select the strata (clusters) to be sampled (first stage) and then uses either random or systematic methods to select elements from each identified cluster (second stage). The cluster sampling method also allows the possibility of selecting all of the elements from each identified cluster. Either way, the subsamples are summarized separately and the information then combined.

NON-PROBABILITY SAMPLING

In nonprobability sampling, you select the items or individuals without knowing their probabilities of selection. Because of this, the theory of statistical inference that has been developed for probability sampling cannot be applied to nonprobability samples.

The two types of nonprobability samples most commonly used are

- Convenience Sampling
- Judgment Sampling

CONVENIENCE SAMPLING A common type of nonprobability sampling is convenience sampling. In convenience sampling, items selected are easy, inexpensive, or convenient to sample. For example, if you were sampling tires stacked in a warehouse, it would be much more convenient to sample tires at the top of a stack than tires at the bottom of a stack. In many cases, participants in the sample select themselves. For example, many companies conduct surveys by giving visitors to their website the opportunity to complete survey forms and submit them electronically. The responses to these surveys can provide large amounts of data quickly and inexpensively, but the sample consists of self-selected web users.

JUDGMENT SAMPLING For many studies, only a nonprobability sample such as a judgment sample is available. In a judgment sample, you get the opinions of preselected experts in the subject matter. Although the experts may be well informed, you cannot generalize their results to the population.

NOTE Nonprobability samples can have certain advantages, such as convenience, speed, and low cost. However, their lack of accuracy due to selection bias and the fact that the results cannot be used for statistical inference more than offset these advantages.

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